

(a) The number of occurrences greater than one for each letter of INCLUSIONEXCLUSION, you get

- I = 3
- N = 3
- C = 2
- L = 2
- U = 2
- S = 2
- O = 2

To find the number of arrangements with each duplicate letter not being unique, we can do

$$\# \text{ arrangements without uniqueness} = \frac{\text{total arrangements w/ each letter being unique}}{\# \text{ of permutations of each duplicate letter}}$$

But another way of thinking about this is just

1. choose 3 out of 18 spots to place the I's
2. choose 3 out of the 15 spots to place the N's
3. choose 2 out of 13 spots to place the C's

and so on

which when you simplify this, it's just

$$\frac{18!}{3! \cdot 3! \cdot 2! \cdot 2! \cdot 2! \cdot 2! \cdot 2!}$$

$$= \boxed{\frac{18!}{(3!)^2 \cdot (2!)^5}}$$

(b) There are two types of orientation moves:

1. roll (3 symmetries)
2. yaw / pitch (2 symmetries)

This means that there's **6 symmetries** ($3 \cdot 2 = 6$)

So let's just count the number of ways of coloring the prism and ignore the symmetries

There are essentially 9 slots and 9 options. Since they must be different, this is just a **permutation**

$$\begin{aligned} \# \text{ of coloring ways} &= {}_9P_9 \\ &= 9! \end{aligned}$$

We can just divide the symmetries with the total ways of coloring

$$\boxed{\frac{9!}{6}}$$

(c) With the constraints that order of the games doesn't matter and each player plays only one game, we can just think of this as

1. choose 6 out of 18 to play one game
2. choose 6 out of 12 to play another game
3. choose 6 out of 6 to play the last game
4. divide by ways of permuting 3 games (since order doesn't matter)

and then for each game, we need to do

1. choose 3 out of the 6 in the game to be on the first team
2. choose 3 out of the remaining 3 to be on the other team
3. divide by ways of permuting 2 teams (since order doesn't matter)

So mathematically this is

$$\begin{aligned}
 & \frac{\binom{18}{6} \cdot \binom{6}{3} \cdot \binom{3}{3}}{2P_2} \cdot \frac{\binom{12}{6} \cdot \binom{6}{3} \cdot \binom{3}{3}}{2P_2} \cdot \frac{\binom{6}{6} \cdot \binom{6}{3} \cdot \binom{3}{3}}{2P_2} \cdot \frac{1}{3P_3} \\
 &= \binom{18}{6} \cdot \binom{12}{6} \cdot \left(\binom{6}{3} \cdot \binom{3}{3} \right)^3 \cdot \binom{6}{6} \cdot \frac{1}{3! \cdot 2! \cdot 2! \cdot 2!} \\
 &= \binom{18}{6} \cdot \binom{12}{6} \cdot \binom{6}{3}^3 \cdot \frac{1}{3 \cdot 2 \cdot 2 \cdot 2 \cdot 2} \\
 &= \boxed{\binom{18}{6} \cdot \binom{12}{6} \cdot \binom{6}{3}^3 \cdot \frac{1}{48}}
 \end{aligned}$$